Lecture 8: Energy Balances on Process Equipment

CBE 20260 / Thermodynamics I

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1 Introduction

In the previous lecture, we derived the general **Open System Energy Balance**. Today, we will apply this equation to the most common unit operations found in chemical plants.

Our goal is to simplify the general equation for specific pieces of equipment to answer engineering questions like: “How much power does this pump require?” or “How much heat must be removed from this reactor?”

1.1 The General Balance Equation

Recall the general form (neglecting potential and kinetic energy for a moment):

$$\frac{d(M\hat{U})}{dt} = \sum_{in} \dot{m}_{in}\hat{H}_{in} - \sum_{out} \dot{m}_{out}\hat{H}_{out} + \dot{Q} + \dot{W}_s + \dot{W}_{ec}$$

For steady-state processes ($\frac{d}{dt} = 0$) with one inlet and one outlet and assuming constant volume ($\dot{W}_{ec} = 0$), this simplifies to:

$$0 = \dot{m}(\hat{H}_{in} - \hat{H}_{out}) + \dot{Q} + \dot{W}_s \tag{1}$$

This general expression is often useful for open system energy balances on equipment, but beware of the underlying assumptions!

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2 Process Flow Diagram Notation

Chemical plants are often represented using **Process Flow Diagrams (PFDs)**. These diagrams use standardized symbols to represent different types of equipment. Here are some common symbols:

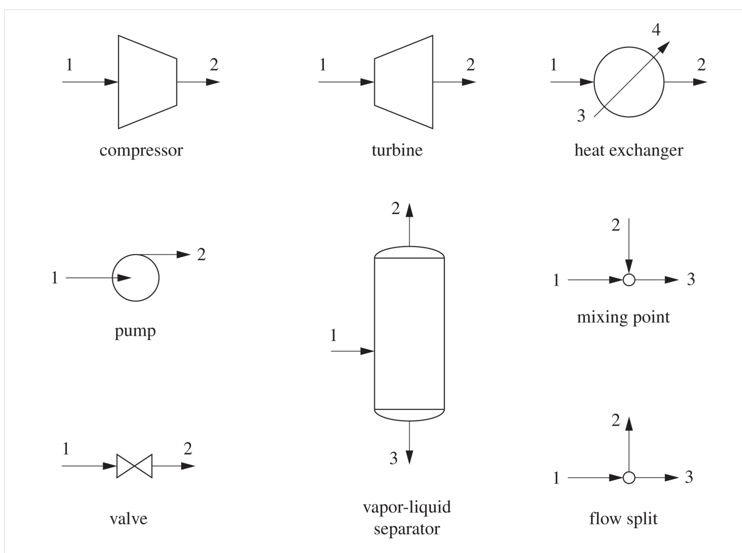

Fig. 1: Typical symbols used to represent different pieces of equipment on a process flow diagram (PFD).

3 Fluid Movers: Pumps and Compressors

These devices are designed to increase the pressure of a fluid.

3.1 Pumps

Pumps are used to move **liquids**. There are two main categories:

1. **Centrifugal Pumps:** Use rotating blades (impellers) to increase fluid velocity, which is converted to pressure.

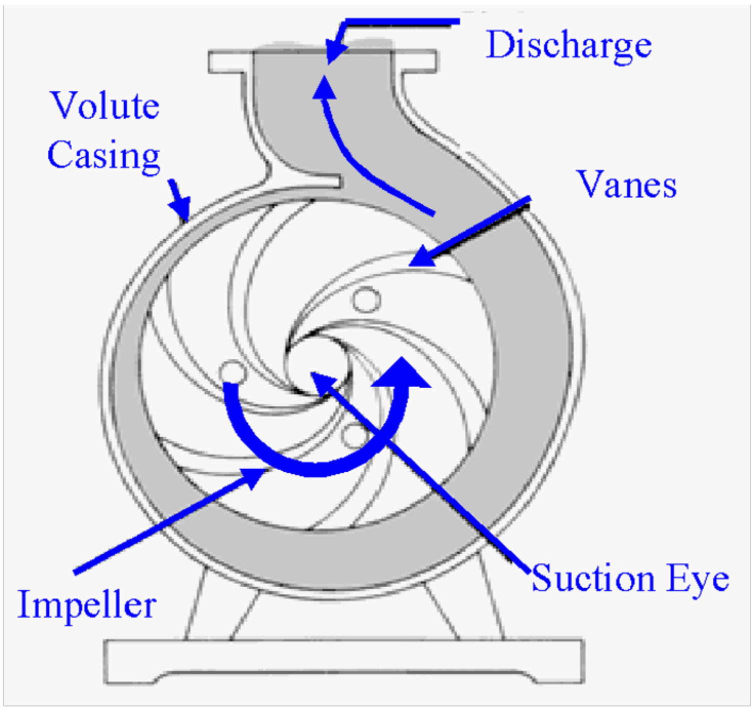

Fig. 2: A schematic of how a centrifugal pump works. Liquid enters at the suction eye, the spinning impeller pushes the liquid forward, increasing its pressure before it is expelled out the discharge. Image from Christopher Okafor.

2. **Positive Displacement Pumps:** Fluid fills a cavity and is physically pushed out (e.g., a piston or gear).

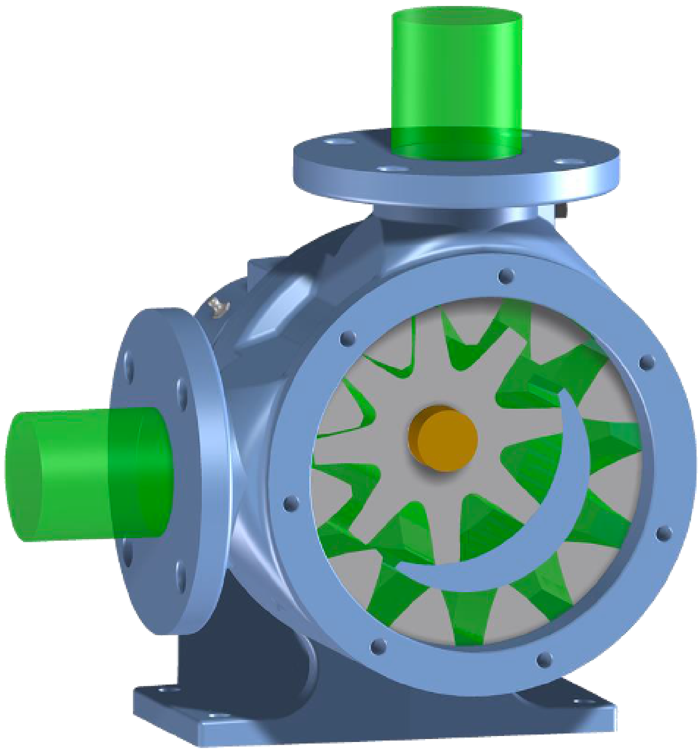

Fig. 3: A schematic of how a positive displacement pump works. Image from Varisco Pumps.

3.1.1 Energy Balance for a Pump

Let's derive the working equation for a pump.

Assumptions:

1. **Steady State**
2. **Adiabatic** ($Q \approx 0$): The fluid moves too fast to lose significant heat.
3. **Negligible Kinetic/Potential Energy changes**: The rise in pressure is the dominant term.

The Balance:

Eq. 1 becomes:

$$0 = \dot{m}(\hat{H}_{in} - \hat{H}_{out}) + \dot{W}_s$$
$$\dot{W}_s = \dot{m}(\hat{H}_{out} - \hat{H}_{in}) = \dot{m}\Delta\hat{H} \tag{2}$$

3.1.2 The Mechanical Definition of Pump Work ($\int VdP$)

While Eq. 2 is correct, it requires us to know the enthalpy at the outlet pressure. Often, we don't know T_{out}, so we can't look up $\hat{H}_{out}$. Even if we did know T_{out}, there aren't tabulated "steam tables" for all fluids. We need an equation purely in terms of P and V.

Alternative Derivation: The Fluid Element Approach

Instead of defining the pump as the system, let's imagine that the system as a small, differential "slug" of fluid as it travels from the pump inlet (State 1) to the outlet (State 2).

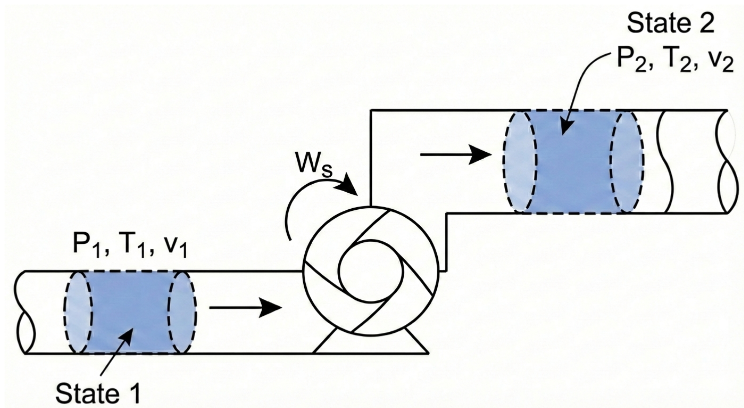

Fig. 4: A fluid packet enters a pump at state 1 and exits at state 2. This is a closed system.

System: A packet of fluid of mass m.

System Type: Closed System (no mass crosses the packet's boundary as it moves).

The governing equation for a closed system is:

$$\Delta[m(\hat{U} + \frac{v^2}{2} + gh)] = W + Q$$

We make the standard assumptions for a pump:

Adiabatic: $Q = 0$.

Negligible Kinetic/Potential Energy changes: $\Delta KE \approx 0, \Delta PE \approx 0$.

This simplifies the balance to:

$$m(\hat{U}_2 - \hat{U}_1) = W_{ec} \tag{3}$$

Notice the difference between Eq. 2 and Eq. 3. They are both correct, but Eq. 2 applies to the pump as a whole system, while Eq. 3 applies to a small fluid element within the pump. In Eq. 3, W_{ec} is the expansion/contraction work done on the fluid packet as it changes volume while moving through the pressure gradient. Note: the system (the fluid element) has no shaft work involved (there is no power source connected to it), so $W_s = 0$. This is sometimes confusing to students, but remember that the pump does shaft work on the fluid as a whole system, while the fluid element only experiences expansion/contraction work. In this derivation, the pump is not the system!

By definition:

$$W_{ec} = -\int_{V_1}^{V_2} PdV$$

So Eq. 3 becomes:

$$\hat{U}_2 - \hat{U}_1 = -\int_{\hat{V}_1}^{\hat{V}_2} Pd\hat{V} \tag{4}$$

Integration by Parts

We usually want the integral Eq. 4 in terms of dP (pressure change), not $d\hat{V}$. We can use integration by parts:

$$\int Pd\hat{V} = P\hat{V} - \int \hat{V}dP$$

Substituting this into our equation:

$$\hat{U}_2 - \hat{U}_1 = -\left[(P\hat{V})\Big|_1^2 - \int_{P_1}^{P_2} \hat{V}dP \right]$$

$$\hat{U}_2 - \hat{U}_1 = -(P_2\hat{V}_2 - P_1\hat{V}_1) + \int_{P_1}^{P_2} \hat{V}dP$$

Rearranging to group the $P\hat{V}$ terms with $\hat{U}$:

$$(\hat{U}_2 + P_2\hat{V}_2) - (\hat{U}_1 + P_1\hat{V}_1) = \int_{P_1}^{P_2} \hat{V}dP$$

Recall the definition of Enthalpy ($\hat{H} \equiv \hat{U} + P\hat{V}$). The left side is simply $\Delta\hat{H}$:

$$\hat{H}_2 - \hat{H}_1 = \int_{P_1}^{P_2} \hat{V}dP \tag{5}$$

Link to Pump Work

Now, recall the open system balance we wrote for the pump itself (see Eq. 2):

$$\hat{W}_s = \Delta\hat{H}$$

By combining Eq. 5 and Eq. 2, we arrive at the mechanical definition of pump work:

$$\hat{W}_s = \int_{P_1}^{P_2} \hat{V} dP \tag{6}$$

Eq. 6 is the general expression we can use to compute the work required by a pump to change a fluid pressure from P_1 to P_2. It is an important result; the work required by a pump is determined by the specific volume of the fluid and the pressure rise.

> **Why Pumps are "Cheap" to Run**
>
> For a liquid, the specific volume $\hat{V}$ is essentially constant (incompressible). We can pull it out of the integral:
>
> $$\hat{W}_{pump} \approx \hat{V}(P_{out} - P_{in})$$
>
> Because liquid volumes are very small ($\hat{V}_{liq} \approx 0.001$ m^3/kg for water), the work required to pressurize a liquid is **small**. We will see that Eq. 6 also applies to compressors, which pressurize gases. Because gas volumes are so much larger than liquids, compressors take a lot more work to compress a gas to the same pressure as a pump requires to compress a liquid.

3.2 Compressors

Compressors are used to pressurize **gases**.

Fig. 5: Top view of a compressor interior showing carefully machined converging blades designed to increase gas pressure from inlet to outlet. Image attribution: Siemens.

The energy balance is formally the same as for a pump:

$$\hat{W}_s = \int_{P_{in}}^{P_{out}} \hat{V} dP$$

However, for a gas, $\hat{V}$ is **not** constant. As pressure rises, volume shrinks drastically.

- Because $\hat{V}_{gas}$ is roughly $1000 \times \hat{V}_{liq}$, the work required to compress a gas is **huge** compared to pumping a liquid to the same pressure.
- To solve the integral $\int \hat{V} dP$, we need an Equation of State (like Ideal Gas or van der Waals) to relate V to P.
- If we have something like the steam tables, we can look up $\hat{H}$ at the inlet and outlet conditions and use Eq. 2 directly. Since $P_2 > P_1$, $\Delta \hat{H}$ is positive, resulting in positive work (work done *on* the system).

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4 Turbines (Expanders)

Turbines are essentially compressors running in reverse. We pass a high-pressure fluid (like steam or gas) through blades to extract shaft work ($\dot{W}_s < 0$).

Fig. 6: View of the interior of a turbine in the Notre Dame power plant. Note the similarity to a compressor. Image attribution: Ed Maginn.

Energy Balance Assume Steady state, adiabatic $Q \approx 0$, and negligible kinetic/potential energy changes.

Note: Kinetic energy changes can sometimes be significant, but we often neglect them for a first pass.

The Balance:

$$\dot{W}_s = \dot{m}(\hat{H}_{out} - \hat{H}_{in}) = \dot{m}\Delta\hat{H}$$

Since $P_{out} < P_{in}$, $\Delta\hat{H}$ is negative, resulting in negative work (work done *by* the system).

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5 Heat Exchangers

Heat exchangers transfer thermal energy between two fluids without mixing them. Common types:

Shell and Tube: One fluid in tubes, another in the surrounding shell. Heat gets transferred through the tube walls between the hot and cold fluids.

Fig. 7: An image of a shell and tube heat exchanger. Image attribution: HRS Heat Exchangers.

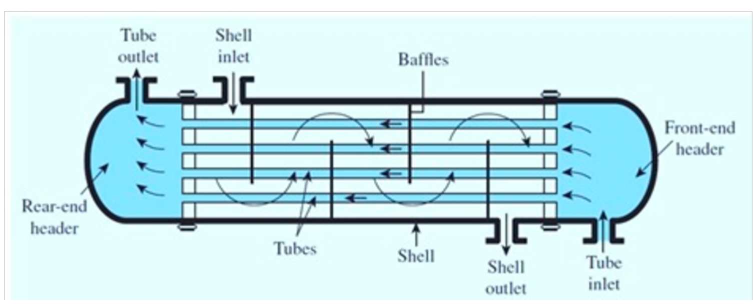

Fig. 8: The inner flow pattern of a shell and tube heat exchanger. Image attribution: Scifed Publishers,

Plate and Frame: High surface area plates separating fluids.

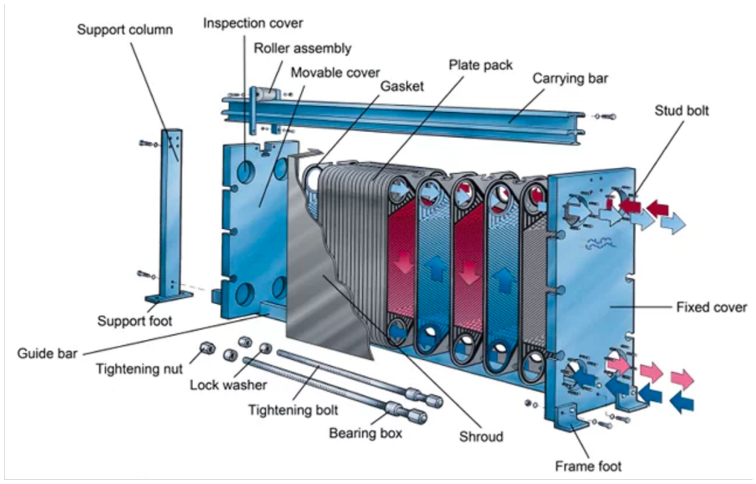

Fig. 9: A plate and frame heat exchanger. Image attribution: Alfa Laval.

Energy Balance Heat exchangers typically have two fluid streams: a hot stream being cooled and a cold stream being heated.

System: We can define the system as just *one* of the fluid streams (e.g., the cold stream being heated).

Assumptions: Steady state and passive ($W_s = 0$).

The Balance:

$$0 = \dot{m}(\hat{H}_{in} - \hat{H}_{out}) + \dot{Q}$$
$$\dot{Q} = \dot{m}(\hat{H}_{out} - \hat{H}_{in}) = \dot{m}C_p\Delta T$$

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6 Example Problem: Steam Turbine

Steam enters a turbine at a mass flow rate of $\dot{m} = 10$ kg/s.

- **Inlet (State 1):** $P_1 = 100$ bar, $T_1 = 500$°C.
- **Outlet (State 2):** $P_2 = 1$ bar, Saturated Vapor.

Calculate the power output ($\dot{W}_s$) of the turbine.

Strategy:

1. Define System: The Turbine.
2. Apply Energy Balance: $\dot{W}_s = \dot{m}(\hat{H}_2 - \hat{H}_1)$.
3. Find State Properties (using Steam Tables).

Solution:

Step 1: Inlet State At 100 bar and 500°C (Superheated Steam): (Looking up in Steam Tables) $\hat{H}_1 = 3373.7$ kJ/kg

Step 2: Outlet State At 1 bar, Saturated Vapor ($x = 1$): (Looking up in Steam Tables) $\hat{H}_2 = 2675.5$ kJ/kg

Step 3: Calculate Work

$$\dot{W}_s = 10\frac{\text{kg}}{\text{s}} \times (2675.5 - 3373.7)\frac{\text{kJ}}{\text{kg}}$$

$$\dot{W}_s = 10 \times (-698.2) \text{ kJ/s}$$

$$\dot{W}_s = -6982 \text{ kW} \approx \mathbf{-7.0} \text{ MW}$$

Result: The turbine produces 7 MW of power.

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7 Comparative Examples: Pumping vs. Compressing

To truly appreciate the difference between liquids and gases, let's calculate the work required to pressurize water versus steam by the same amount.

7.1 Example 1: Pumping Water

Problem: Calculate the specific shaft work ($\hat{W}_s$) required to pump saturated liquid water from $P_1 = 0.5$ bar to $P_2 = 5$ bar.

State 1 (Inlet):

- $P_1 = 0.5$ bar, Saturated Liquid
- From Steam Tables:
  - $T_1 = 81.32$°C
  - $\hat{V}_1 = 0.001030 \text{ m}^3/\text{kg}$
  - $\hat{H}_1 = 340.5$ kJ/kg

⚠ Pitfall: Using Enthalpy Tables for Liquids

You might be tempted to use the open system balance $\hat{W}_s = \Delta\hat{H}$ and look up $\hat{H}_2$ in the subcooled liquid tables. At $P_2 = 5$ bar, the temperature will only rise slightly (approx 81.32°C). If we look at a subcooled water table for 5 bar and 80°C, we find $\hat{H} \approx 335.4$ kJ/kg. If we calculate work:

$$\hat{W}_s = 335.4 - 340.5 = -5.1 \text{ kJ/kg}$$

This is impossible! It implies the pump *generates* power. Enthalpy is insensitive to pressure for liquids, so reading small differences from tables leads to huge errors.

Correct Approach: The Mechanical Definition For a liquid, we assume it is incompressible ($\hat{V} \approx$ constant).

$$\hat{W}_s = \int_{P_1}^{P_2} \hat{V} dP \approx \hat{V}_1 (P_2 - P_1)$$

$$\hat{W}_s = 0.001030 \frac{\text{m}^3}{\text{kg}} \times (5 - 0.5) \text{ bar}$$

Note: We must convert bar to Pascals (N/m^2) to get Joules.

$$\hat{W}_s = 0.001030 \times (4.5 \times 10^5 \text{ Pa})$$

$$\hat{W}_s = 463.5 \text{ J/kg}$$

Result: The work required is **463.5 J/kg** (or 0.46 kJ/kg).

7.2 Example 2: Compressing Steam

Problem: Now, calculate the work required to compress steam (vapor) from the same starting pressure (0.5 bar) to the same final pressure (5 bar). Assume the inlet is superheated at 300°C.

State 1 (Inlet):

- $P_1 = 0.5$ bar, $T_1 = 300°\text{C}$
- $\hat{H}_1 = 3075.2$ kJ/kg

State 2 (Outlet):

- $P_2 = 5$ bar.
- Determining the exact outlet temperature requires concepts involving *entropy* (which we will cover later). For now, we utilize the condition for minimum theoretical work (isentropic). That is, I have set this problem up so that the calculation gives the *minimum* amount of work required to compress the steam.
- Under these conditions, the outlet enthalpy is found to be $\hat{H}_2 = 3927$ kJ/kg.

Calculation: Since we are dealing with a gas, we *can* use the enthalpy difference directly (changes in H are large for gases).

$$\hat{W}_s = \hat{H}_2 - \hat{H}_1$$

$$\hat{W}_s = 3927 - 3075.2 = 851.8 \text{ kJ/kg}$$

$$\hat{W}_s = 851,800 \text{ J/kg}$$

7.3 Comparisons and Conclusion

Compare the work required for the two fluids:

1. **Pump (Liquid):** ~464 J/kg
2. **Compressor (Gas):** ~851,800 J/kg

$$\frac{\text{Work}_{gas}}{\text{Work}_{liq}} \approx 1836$$

Why such a huge difference? Look back at the mechanical work equation: $\hat{W}_s = \int \hat{V} dP$. The work is directly proportional to the specific volume $\hat{V}$. * $\hat{V}_{liquid} \approx 0.001 \text{ m}^3/\text{kg}$ * $\hat{V}_{gas} \approx 1.0 \text{ m}^3/\text{kg}$

Because the gas volume is roughly $1000\times$ larger than the liquid volume, it costs roughly $1000\times$ more energy to pressurize it. This is why pumps are cheap to run, but gas compressors are massive energy consumers in a chemical plant. This is an important concept to understand as we design processes! It will become important when we discuss power generation and refrigeration cycles later in the course.